A Uniformly Discrete Coset-Quotient Subcase of a Question of Cuth–Doucha

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Source Question

This packet concerns Question 5.5 from Marek Cuth and Michal Doucha, *Projections in Lipschitz-free spaces induced by group actions*, arXiv:2104.11519 [1].

$\mathrm{Lip}_0(X, d)$. Thus, a possible way of providing a positive answer to Question 5.4 would be to show that $\mathrm{Lip}_0(X, d) \overset{C}{\hookrightarrow} \mathrm{Lip}_0(X, \|\cdot\|)$.

In our proofs we were using the assumption that G has bounded orbits. In this case one could ask e.g. the following which aims at pushing forward what was initiated in Theorem 4.5.

Question 5.5. Let G be a metric group which is amenable and locally compact or SIN. Let $H \subset G$ be its bounded subgroup. Is it true that $\mathcal{F}(G/H) \overset{C}{\hookrightarrow} \mathcal{F}(G)$?

In transcription, the question asks:

> Let G be a metric group which is amenable and locally compact or SIN. Let $H \subset G$ be its bounded subgroup. Is it true that $\mathcal{F}(G/H) \overset{C}{\hookrightarrow} \mathcal{F}(G)$?

Claimed Status

Candidate partial result, likely valid pending human review. We prove the subcase in which the coset quotient metric is uniformly discrete. The general question remains open.

Definitions and Setup

Let (G, d) be a group equipped with a left-invariant metric and let H be a subgroup. Write e for the identity element and set

$$B = \sup_{h \in H} d(e, h).$$

Thus H is bounded exactly when $B < \infty$.

On the left-coset space G/H consider

$$D(gH, fH) = \inf_{h,k \in H} d(gh, fk).$$

We assume that D is a metric, and that it is uniformly discrete: there is a number $a > 0$ such that $D(u, v) \geq a$ whenever $u \neq v$ in G/H. The quotient map is denoted by $q : G \to G/H$.

Proof Intuition

The obstruction in the noncompact bounded-subgroup problem is that the usual averaging over H need not land back in $\mathcal{F}(G)$; an invariant mean naturally produces bidual objects. In the uniformly discrete quotient case one can bypass averaging completely. Choose one representative in each coset. Because every coset has diameter at most controlled by the bounded subgroup H, this arbitrary transversal is coarse Lipschitz with an additive error $2B$. Uniform discreteness of the quotient converts the additive error into an honest Lipschitz bound. The universal property of Lipschitz-free spaces then linearizes the quotient map and the section, giving a bounded projection.

Lemma 1 (Bounded fibers plus quotient separation give a Lipschitz section)**.** *Let (G, d), H, B, D, and a be as above. Then every set-theoretic section $s : G/H \to G$ of q with $s(H) = e$ is Lipschitz, with*

$$\mathrm{Lip}(s) \leq 1 + \frac{2B}{a}.$$

Proof. Let $u, v \in G/H$. If $u = v$, there is nothing to prove. Suppose $u \neq v$. Choose $\varepsilon > 0$. By the definition of D, there are $h, k \in H$ such that

$$d(s(u)h, s(v)k) \leq D(u, v) + \varepsilon.$$

Since d is left-invariant,

$$d(s(u), s(u)h) = d(e, h) \leq B \quad \text{and} \quad d(s(v)k, s(v)) = d(e, k^{-1}) \leq B.$$

Therefore

$$\begin{aligned} d(s(u), s(v)) &\leq d(s(u), s(u)h) + d(s(u)h, s(v)k) + d(s(v)k, s(v)) \\ &\leq D(u, v) + 2B + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ and using $D(u, v) \geq a$ gives

$$d(s(u), s(v)) \leq \left(1 + \frac{2B}{a}\right) D(u, v).$$

□

Theorem 1. *Under the hypotheses above, $\mathcal{F}(G/H, D)$ is linearly isomorphic to a complemented subspace of $\mathcal{F}(G, d)$. More precisely, there is a projection on $\mathcal{F}(G, d)$ onto a copy of $\mathcal{F}(G/H, D)$ with norm at most $1 + 2B/a$.*

Proof. The quotient map $q : (G, d) \to (G/H, D)$ is 1-Lipschitz, because $D(qg, qf) \leq d(g, f)$ by taking $h = k = e$ in the infimum defining D.

Choose a section $s : G/H \to G$ with $s(H) = e$. By the lemma, s is L-Lipschitz with $L \leq 1 + 2B/a$. Since $q \circ s$ is the identity map on G/H, the induced linear maps on Lipschitz-free spaces satisfy

$$\widehat{q} \circ \widehat{s} = I_{\mathcal{F}(G/H)}.$$

Consequently

$$P = \widehat{s} \circ \widehat{q} : \mathcal{F}(G, d) \to \mathcal{F}(G, d)$$

is an idempotent operator. Its range is $\widehat{s}[\mathcal{F}(G/H, D)]$, and the restriction of $\widehat{q}$ to this range is the inverse of $\widehat{s}$. Thus the range is linearly isomorphic to $\mathcal{F}(G/H, D)$. Moreover

$$\|P\| \leq \|\widehat{s}\| \, \|\widehat{q}\| \leq L \leq 1 + \frac{2B}{a}.$$

$\square$

Verification and Scope

The proof uses only the universal property of Lipschitz-free spaces and the elementary section estimate above. No amenability, local compactness, or SIN assumption is used. The price is the additional hypothesis that the coset quotient metric is uniformly discrete.

This is a genuine partial result for Question 5.5, but it does not touch the small-scale quotient case. If D has nonzero distances tending to zero, the estimate

$$d(s(u), s(v)) \leq D(u, v) + 2B$$

does not imply that s is Lipschitz. That is the remaining obstruction for this route.

Novelty and Literature Check

Before promotion, the run checked the cheap indexes `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` for arXiv:2104.11519 and the core phrases `F(G/H)`, `bounded subgroup`, `Lip_0(X/Y)`, and `Lipschitz-free`. No exact duplicate packet was found.

The local parsed arXiv corpus found the exact question only in arXiv:2104.11519. It also found the related compact-subgroup averaging projection in Doucha–Kaufmann [2], which handles compact H by Haar averaging. The present packet is not a literature identification of that result: it proves a different, elementary uniformly discrete quotient subcase, including cases where no compact averaging argument is available. Web searches for exact phrases around `F(G/H) complemented F(G)` and `Lip_0(X/Y) complemented` did not reveal an exact later answer to the source question.

Human Review Recommendation

Review the metric assumptions first: in particular, check that the intended quotient metric D is a genuine metric and that the quotient has positive separation a. Under those assumptions, the proof reduces to the displayed triangle inequality and the standard linearization argument for Lipschitz-free spaces.

References

[1] M. Cuth and M. Doucha, *Projections in Lipschitz-free spaces induced by group actions*, arXiv:2104.11519, 2021.

[2] M. Doucha and P. L. Kaufmann, *Approximation properties in Lipschitz-free spaces over groups*, arXiv:2005.09785, 2020.

[3] N. Weaver, *Lipschitz Algebras*, World Scientific, 1999.